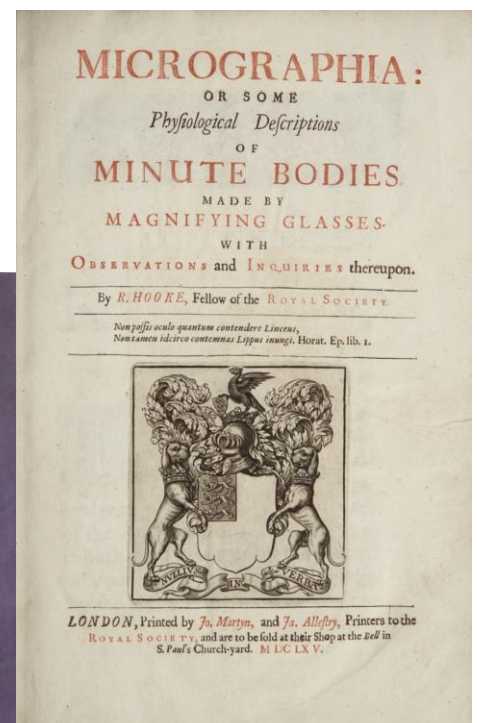
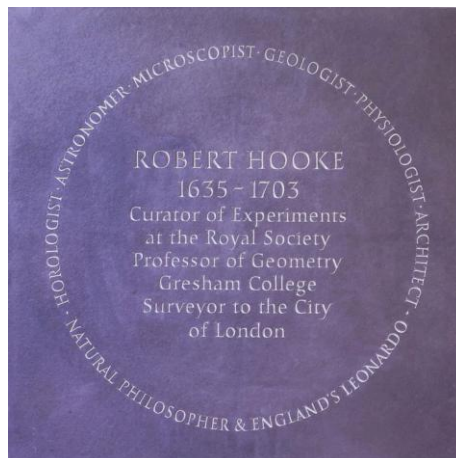


CERAMIC ENGINEERING

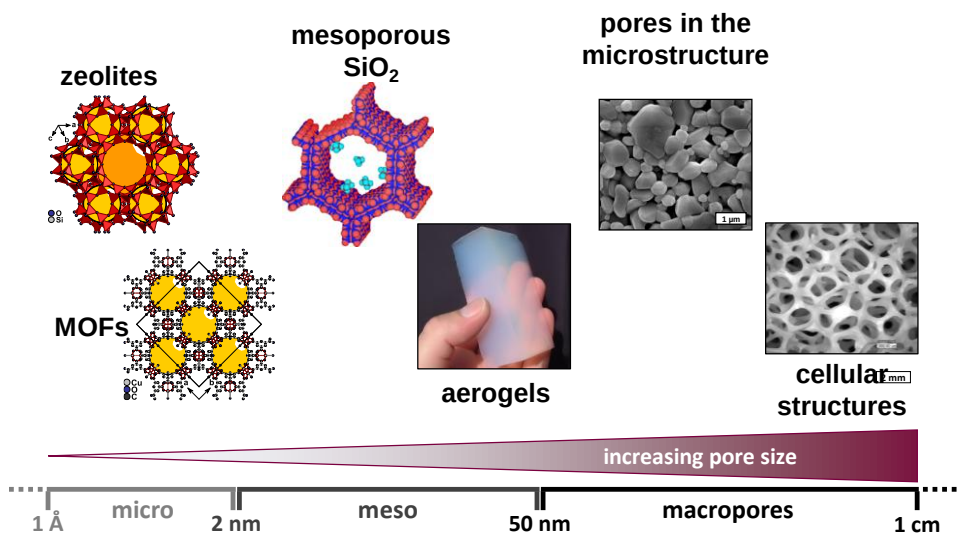
Chapter 8: cellular materials



„I could exceedingly plainly perceive it to be all perforated and porous, much like a Honey-comb, but that the pores of it were not regular...these pores, or cells,...were indeed the first microscopical pores I ever saw, and perhaps, that were ever seen, for I had not met with any Writer or Person, that had made any mention of them before this...”.



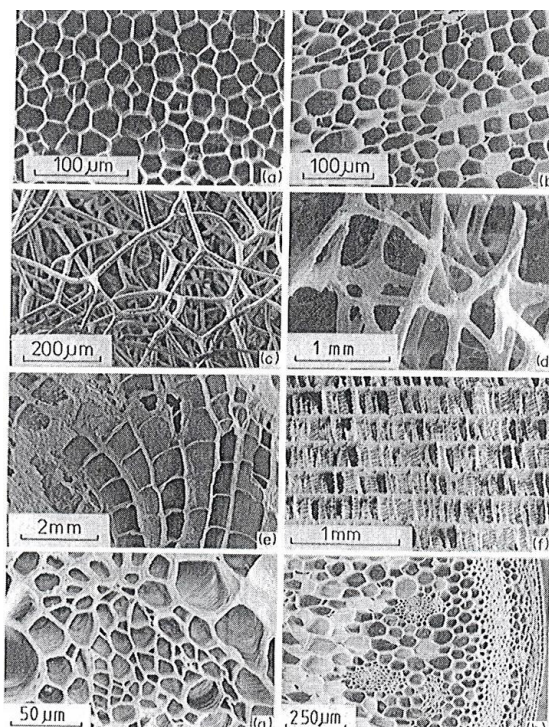
porosity in other materials: zeolites, MOFs, silica gels



M. Thommes et al.,
Pure Appl. Chem. **2015**, 87, 1051.

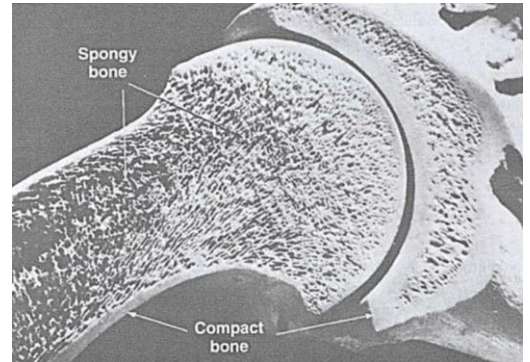
natural cellular materials:

- (a) cork
- (b) balsa
- (c) sponge
- (d) cancellous bone
- (e) coral
- (f) cuttlefish
- (g) iris leaf
- (h) stalk of a plant



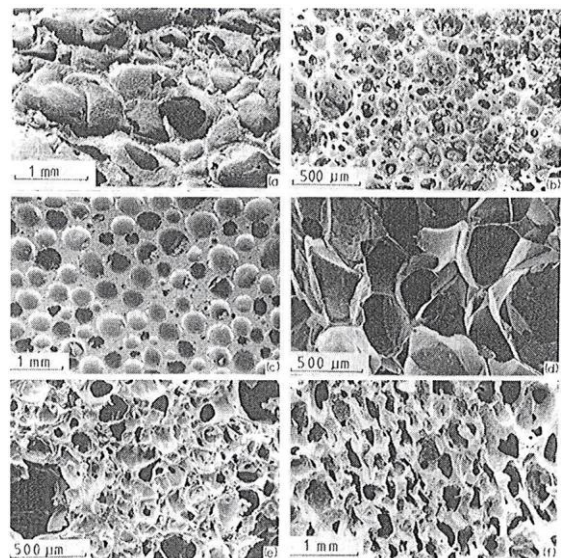
mamal bones:

saggital section of the proximal end of the humerus in relation of the glenoid fossa of the scapula at the shoulder joint (here: dry bones, cartilaginous articular surfaces of the joint not present)



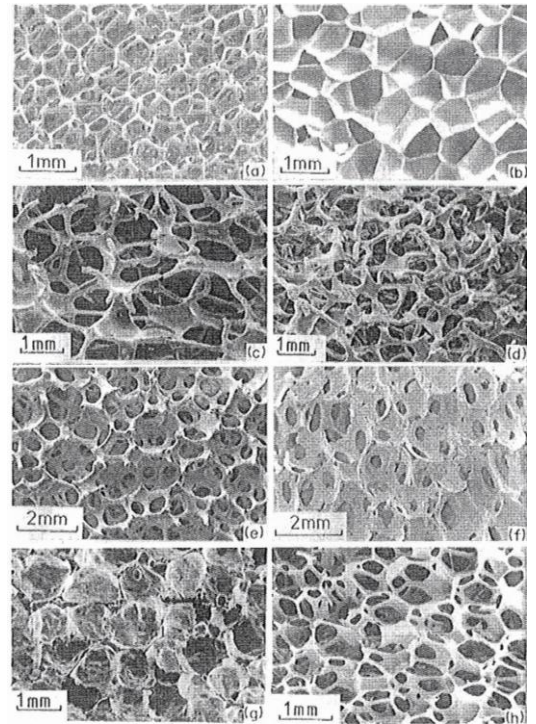
food foams:

- (a) bread
- (b) meringue
- (c) chocolate
- (d) junk food crisp
- (e) malteser
- (f) jaffa cake



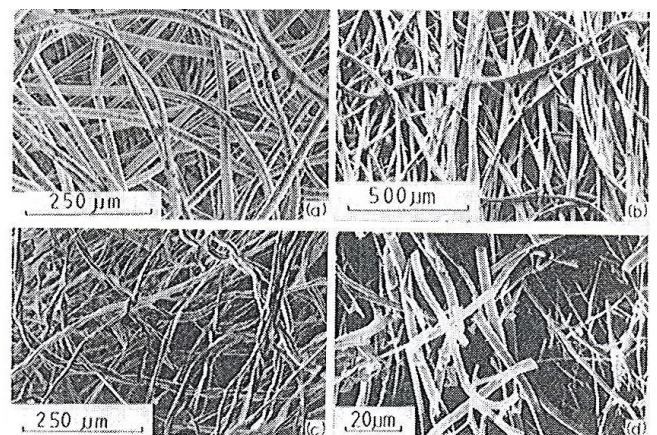
three-dimensional cellular materials:

- (a) open-cell polyurethane
- (b) closed-cell polyethylene
- (c) nickel
- (d) copper
- (e) zirconia
- (f) mullite
- (g) glass
- (h) PE foam with both,
open and closed cells

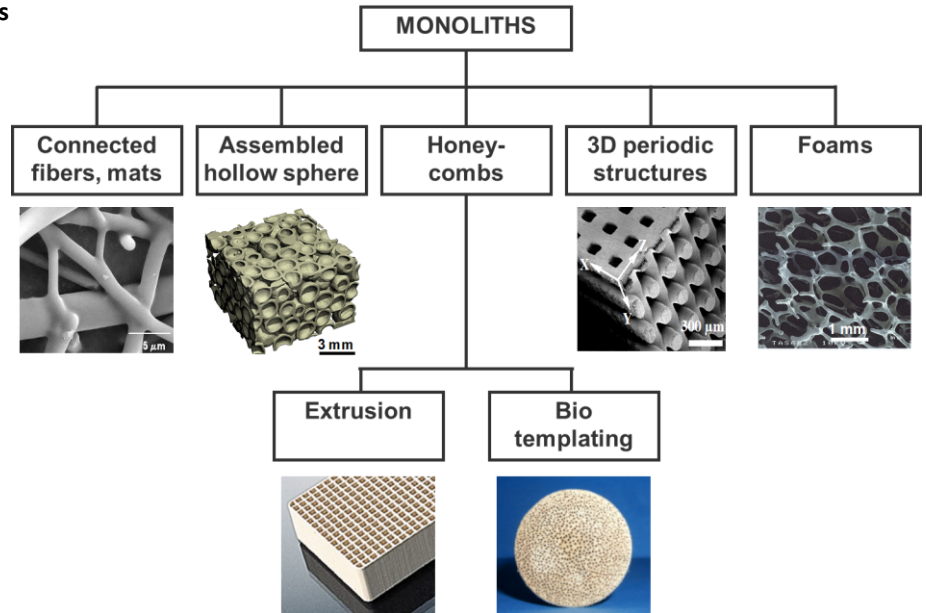


fibrous materials:

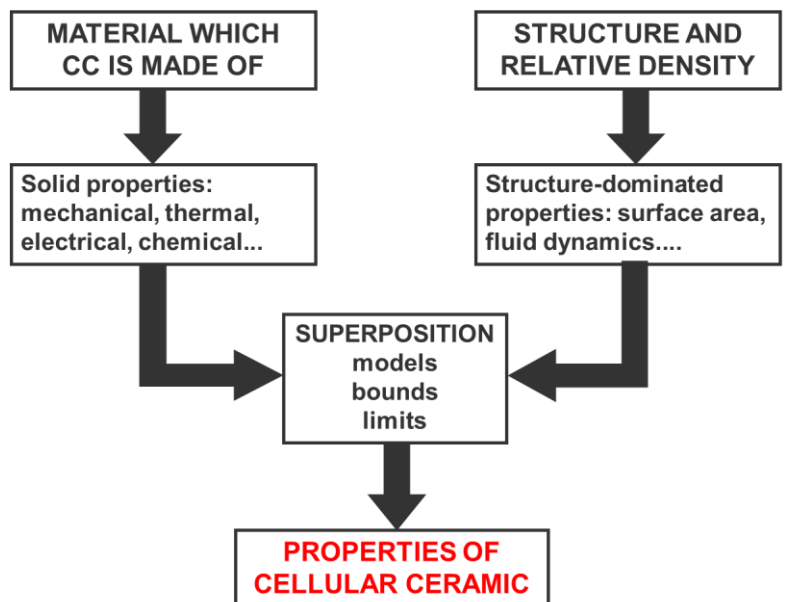
- (a) felt
- (b) paper
- (c) cotton wool
- (d) space shuttle tile



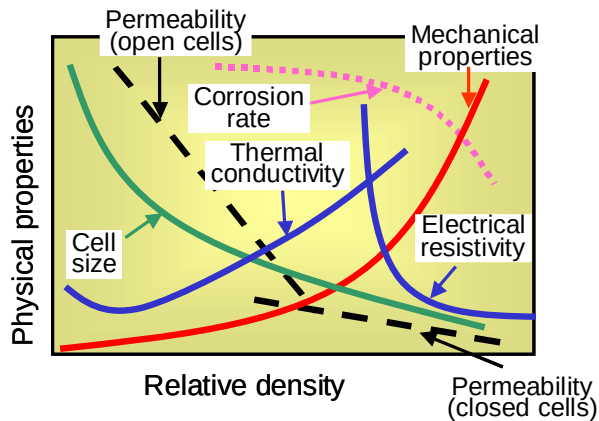
most relevant structures



bulk vs. cellular matter



Properties (general) as a function of porosity



low:

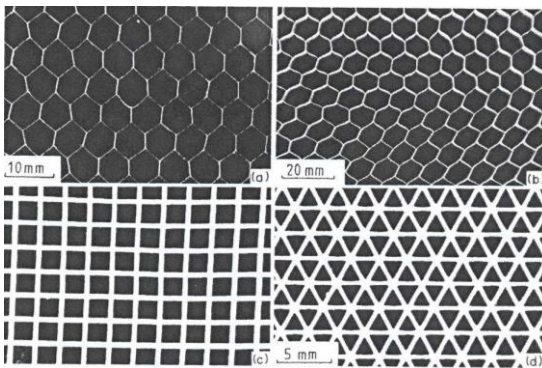
- density
- thermal conductivity
- dielectric constant
- thermal mass

high:

- specific surface area
- thermal shock resistance
- specific strength
- dimensional stability
- permeability

→ cellular ceramics extend the range of properties of ceramic materials

The honeycomb structure(s)

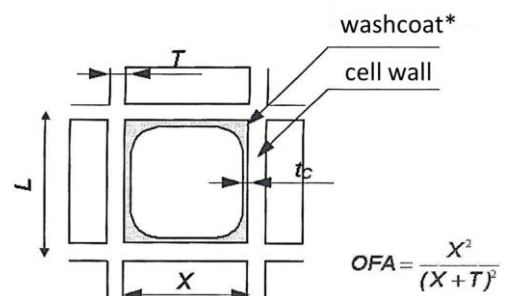


Two-dimensional cellular material:

- aluminium honeycomb,
- paper-phenolic resin honeycomb,
- ceramic honeycomb with square cells,
- Ceramic honeycombs with triangular cells

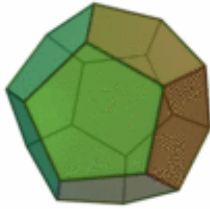
T – wall thickness
 t_c – washcoat thickness
 x – cell width
 L – repeating unit

OFA – open front area



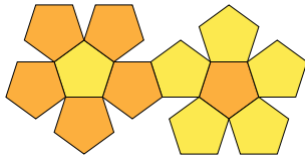
* washcoat, if a catalyst is applied

Rigid foams and their structures



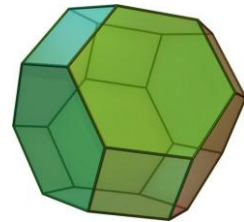
pentagondodecahedron

- no complete room filling
- part of real foams

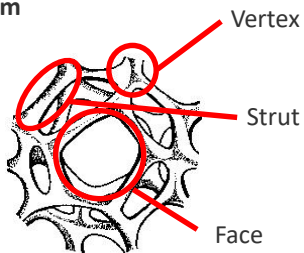


cubo-octahedron

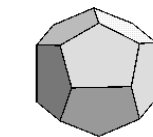
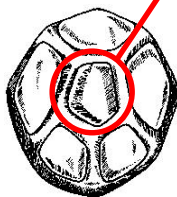
- complete room filling
- part of real foams
- Lord Kelvin: lowest surface energy energy of a polyhedron



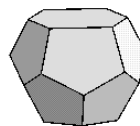
open-cell foam



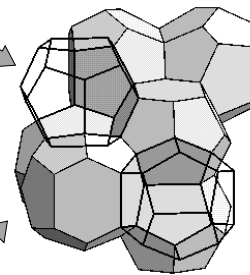
closed-cell foam



2 pentagon-dodecahedra



6 Goldberg-polyhedra

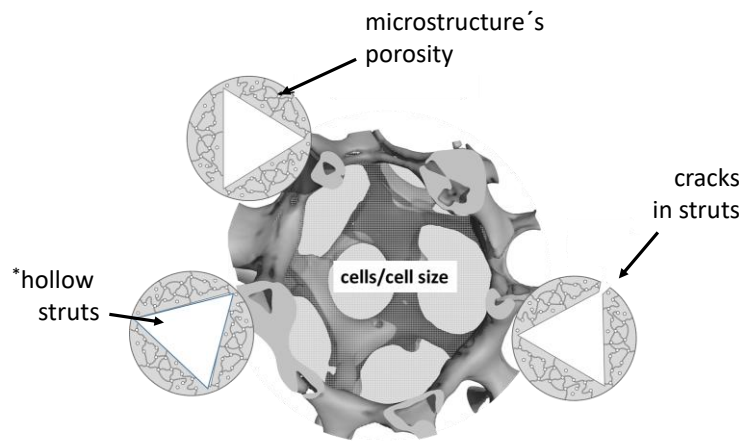


WEAIRE*-PHELAN unit cell:

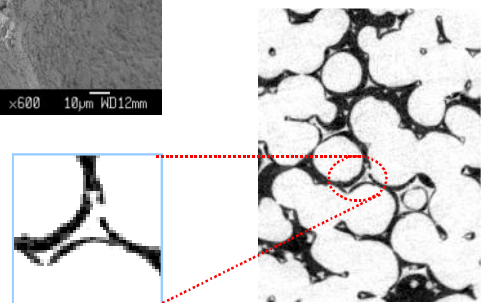
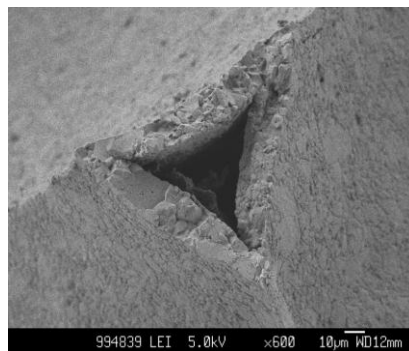
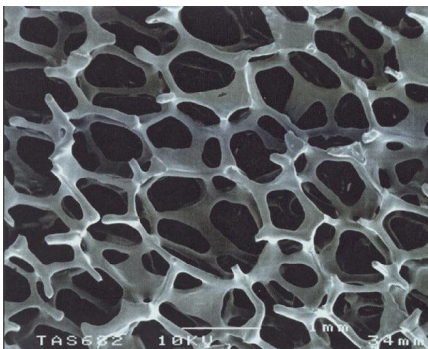
surface energy – 2 % compared to the Kelvin cell

*Denis Weaire, Trinity College Dublin, Ireland, opening speaker
(A Mystery Tour through Cellular Materials)
at the 4th Cellmat, 2016, Dresden

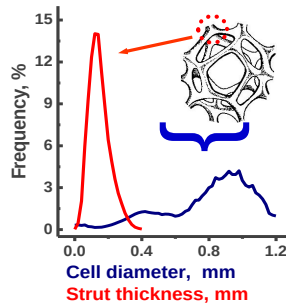
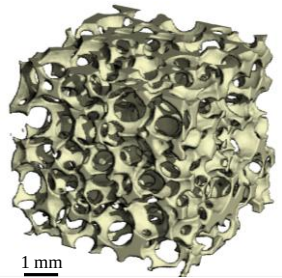
porosity in rigid foams



* foam struts can be solid or hollow, this depends on the fabrication methode
 solid struts by direct foaming; hollow struts by reticulated foaming; however, porosity
 characterization of foams is a must



reticulated ceramic foam by μ -CT



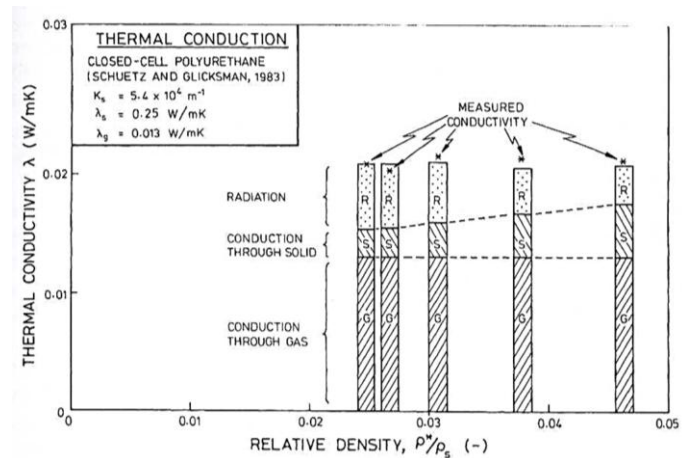
structural properties

- fractional density, % 5 – 40
- mean cell diameter, mm 0.5 – 2.0
- mean strut thickness, mm 0.2 – 0.5
- sphere-like cells/
isotropic cell arrangement

thermal conduction in foams, example PU

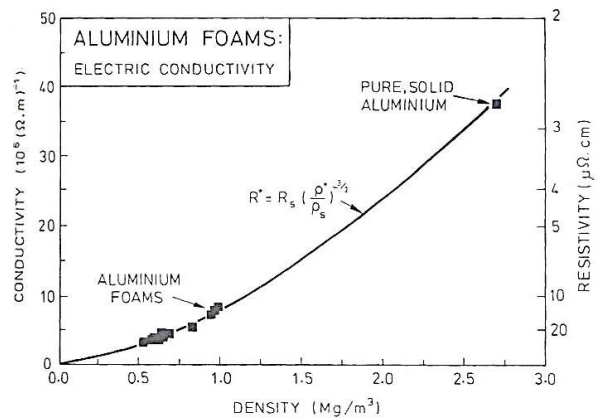
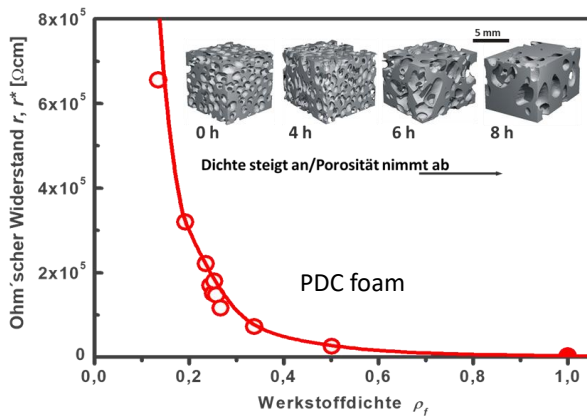
the contributions of conduction through the gas, conduction through the solid and radiation to the thermal conductivity of closed-cell rigid polyurethane foams;

as the relative density of the foams increases, the contribution from conduction increases while that from radiation decreases



electric conductivity in foams, example Al

electric conductivity and resistivity of Al
foams including that of the bulk aluminium



Manufacturing processes of cellular solids

foams:	direct foaming reticulation processing spaceholder techniques Investment/precision casting
honeycombs:	extrusion punching waving/expanding and wrapping template processing
3D period. stuctures	AM methods (printing, direct writing, robocasting...)
fiber mats:	spinning/spinning and partial sintering

example reticulated porous ceramics:

patent pending in 1963

May 21, 1963

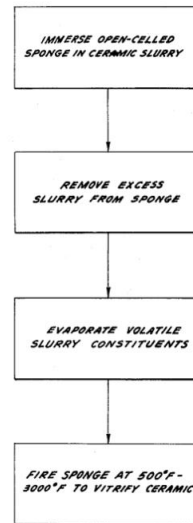
K. SCHWARTZWALDER ET AL 3,090,094

METHOD OF MAKING POROUS CERAMIC ARTICLES

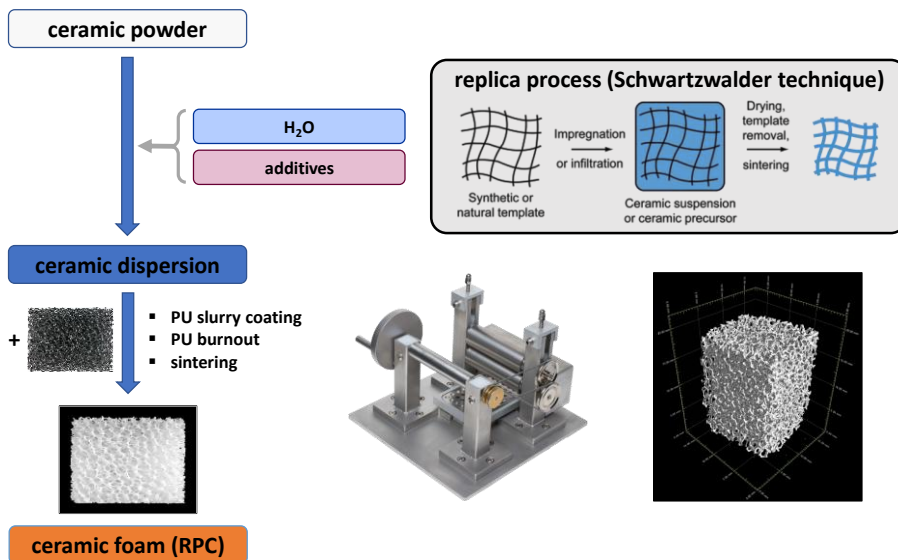
Filed Feb. 21, 1961

Claimed herein:

- template foam with cell size
- binder materials
- ceramic materials
- firing procedure

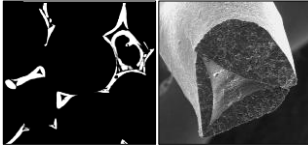


INVENTORS
Karl Schwartzwalder
Walter H. Somers
BY *K. H. Thompson*
ATTORNEY



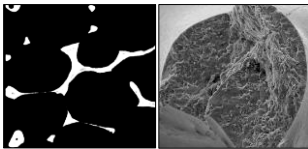
new approach

hollow strut infiltration with Al_2O_3 nano particles, $d \approx 200 \text{ nm}$
 → infiltration without vacuum at ambient pressure

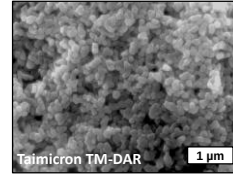


infiltration by:

- immersion
- (vacuum)



infiltration slurry
 $\text{n-Al}_2\text{O}_3$
 50-65 m.%
 in water



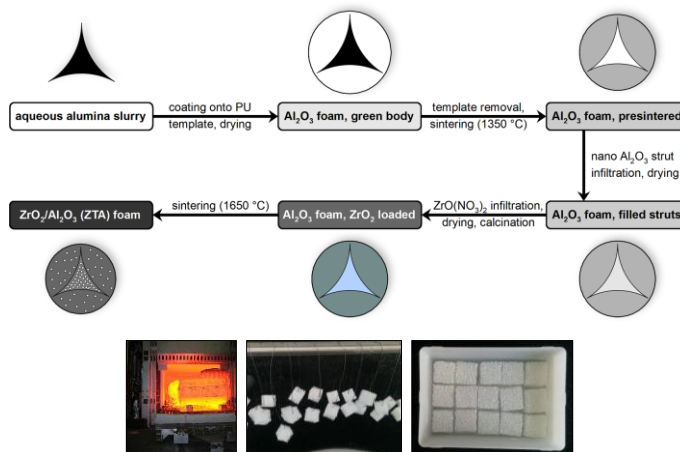
procedure

- 1) presintering of Al_2O_3 RPCs (1350-1650 °C)
- 2) infiltration (ambient pressure or vacuum)
- 3) final sintering of Al_2O_3 foams (1100-1650 °C)



procedure:

- 1) hollow strut infiltration of pre-sintered alumina foams with submicron Al_2O_3
- 2) infiltration of microstructural porosity with $\text{ZrO}(\text{NO}_3)_2$
- 3) calcination: $\text{ZrO}(\text{NO}_3)_2 \rightarrow \text{ZrO}_2$ (600°C) and final sintering (1650 °C)



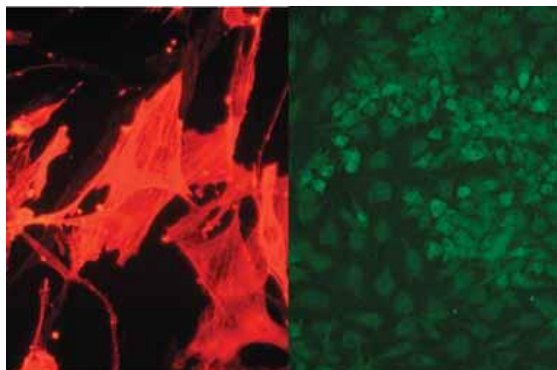
Some applications of cellular materials

- liquid metal filtration
- exhaust gas filtration
- catalyst support
- aeration of liquids
- porous burners
- bone substitute
- material reinforcement

Biomaterials

Bioengineering of Semiautologous Arterial Vessels with Reconstructed Media and Intima, Longtime Tested In Vivo

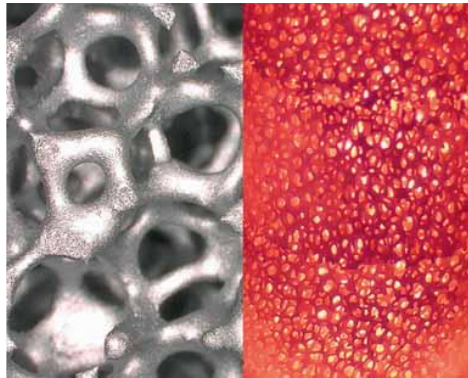
Tolga Atilla Sagban, Eleonore Schiegel, Klaus Grabitz, Wilhelm Sandmann and Kai Michael Balzer



High temperature and energy technology

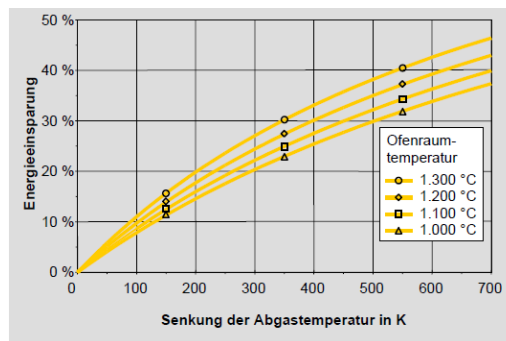
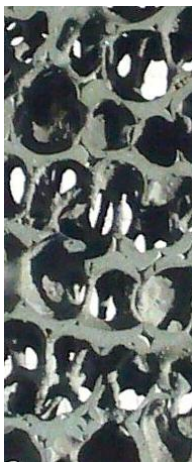
Cellular Ceramics in Combustion Environments

Alexander Füssel, Daniela Böttge, Jörg Adler, Felix Marschallek and Alexander Michaelis

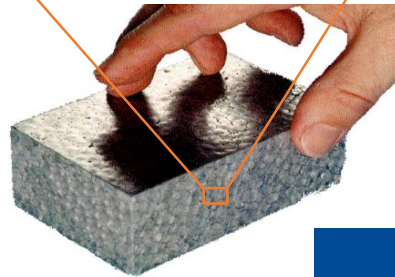
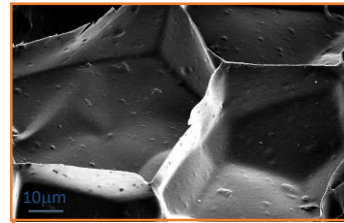
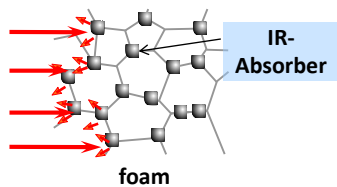
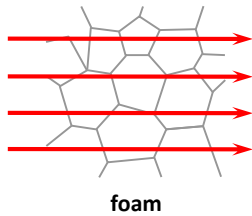


High temperature and energy technology

Heat storage material in high-temperature recuperators



Thermal Insulation



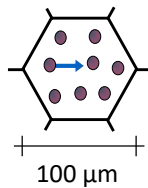
Neopor®



Thermal insulation

macro foams

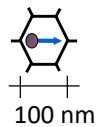
- significant influence on thermal conductivity



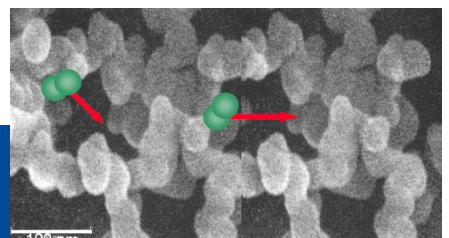
mean free path at 1 bar: 70 nm
Pore size: 20 - 200 μm

nano foams

- no influence on thermal conductivity!



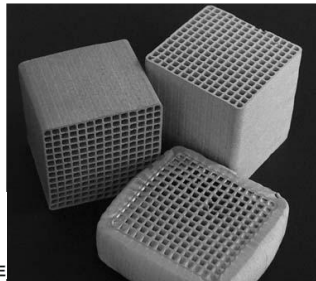
mean free path at 1 bar: 70 nm
Pore size: 200 - 500 nm



Basic research (right)
and transfer (left)

Martensitic Phase Transformation in TRIP-Steel/Mg-PSZ Honeycomb Composite Materials on Mechanical Load

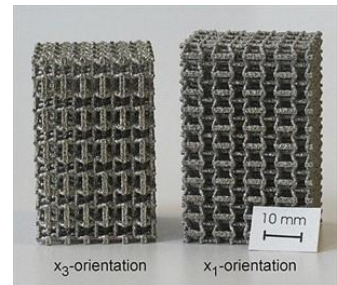
Christian Weigelt, Christos G. Aneziris, Harry Berek, David Ehinger and Ulrich Martin



Design of Auxetic Structures via Mathematical Optimization

J. Schwerdtfeger, F. Wein, G. Leugering, R. F. Singer, C. Körner, M. Stingl, F. Schury

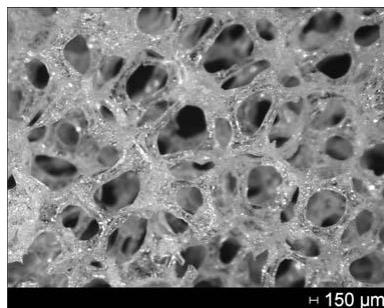
Adv. Mater.



Environmental
catalysis

Preparation of Optically Transparent Open-Celled Foams and its Morphological Characterization Employing Volume Image Analysis

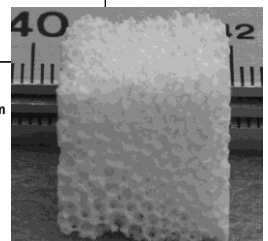
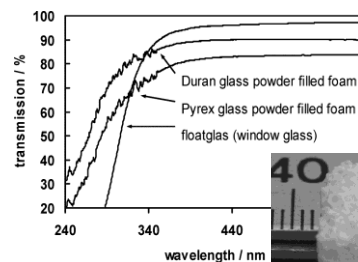
Ethel C. Bucharsky, Karl G. Schell, Peter Habisreuther, Rainer Oberacker, Nikolaos Zarzalis and Michael J. Hoffmann



Novel Open-Cellular Glass Foam with High Optical Transmittance

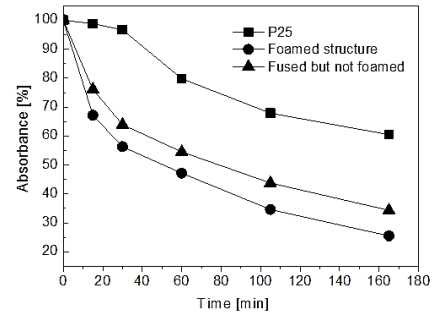
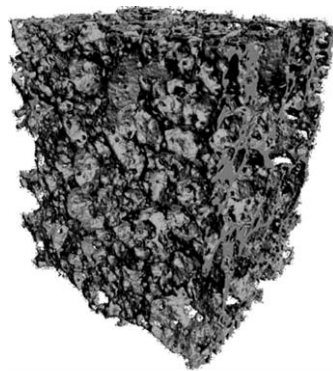
Christiane Ohl¹⁾, Mathias Kappa¹⁾, Viola Wilker¹⁾, Sarama Bhattacharjee²⁾, Franziska Scheffler³⁾ and Michael Scheffler

J. Am. Ceram. Soc. 2011



Photocatalytic applications

Novel-Type Inorganic Foams from Preceramic Polymers with Embedded Titania Nanoparticles for Photo-Catalytic Applications



Sarama Bhattacharjee, Pratik Ranjan Das, Christiane Ohl, Viola Wilker, Mathias Kappa, Franziska Scheffler and Michael Scheffler

Structural light weight applications

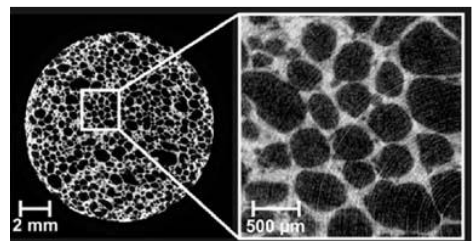
Structural and Material Design of Open-Cell Powder Metallurgical Foams

Peter Quadbeck, Kerstin Kümmel, Ralf Hauser, Gisela Standke, Jörg Adler, Günter Stephani and Bernd Kieback



Aluminum Integral Foams with Near-Microcellular Structure

Johannes Hartmann, André Trepper and Carolin Körner



Structural light weight applications

Analysis of Cellular Metals as Energy-Absorbing Elements in Car Seats

Srecko Nesic, Peter Schäffler, Klaus Unruh, Wilhelm Michels and Ulrich Krupp

